



# CANFARMSAFE™

Act 3 - 02

## Robotic Rock Picking

Due to the short growing season in Canada and the inadequacy of available mechanical rock pickers, potato growers continue removing rocks by hand. A team of researchers at Dalhousie University is currently developing a new technology that will help resolve this issue. The goal of the project is two fold:

- Map the location of rocks while planting the potatoes and
- Automate the process of removing the rocks using a robotic system with a 3D camera and a robotic arm.

### Mapping Rocks

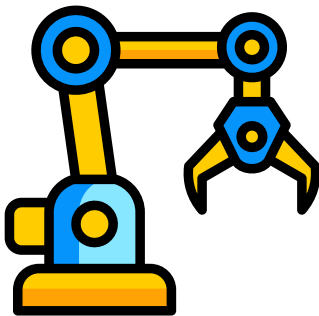
The machine vision system in development will be mounted at the rear of potato planters during regular operation. The vision system will be able to identify, estimate the weight, and tag the locations of large stones to simplify the rock picking process. By integrating the vision system with the GPS system in the tractor, it will be possible to provide a rock distribution map for farm managers.



Since rocks on soil can be identified by naked human eye, it is expected that an image processing technique developed on RGB images will be able to accomplish the task of accurately detecting the rocks. Mounting the camera on the planter will provide an accessible platform to scan the field without the need to bring any extra aerial or ground vehicle to carry the camera. Moreover, integrating the vision system to the GPS system already existing on the planter which limit the cost of the hardware to components needed for imaging only.

## **Robotic Rock Picker**

In the second part of this project, a robotic arm mounted on a cart pulled by a tractor will be designed and programmed to pick up the rocks identified by the machine vision system. Robotic arms are used in factories by many different industries. In agriculture, robots are used to pick produce like tomatoes or eggplants, but special “hands” or end-effectors have also been designed to pick heavy crops such as pumpkins. However, unlike pumpkins which have more or less regular shapes, rocks may have irregular shapes that might add more challenges when designing the end-effector, besides the weight.



A 3D camera will be used to find the rocks and control the robotic arm system. At first, the tractor operator will drive towards a rock until it is located within the field of view of the 3D camera and can be picked up by the robotic arm. As development progresses, the 3D camera will be able to scan a wider area to collect all the rocks within reach at one stop before moving to another location.

Rather than needing to spend time before planting picking rocks out of the fields, this project will enable farmers to plant their crops earlier and will limit the amount of manual rock picking. This technology will therefore reduce the risks of muscle strain, repetitive stress injuries, heat exhaustion, UV exposure, dehydration, or overexertion associated with manual rock picking.



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